Completely conditional-fault edge-bipancyclicity of hypercubes *

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Abstract

In this paper, we consider the conditionally faulty graphs G that each vertex of G is incident with at least m fault-free edges, $2 \leq m \leq \delta(G)-1$. *We extend the limitation* $m \geq 2$ *in all previous results of edge-bipancyclicity with faulty edges and faulty vertices. Let* f_e *(respectively,* f_v*) denotes the number of faulty edges (respectively, faulty vertices) in an n-dimensional hypercube* Q_n. *For all* $m \leq min\{n - \lceil \frac{f_v}{2} \rceil, n-1\}$ *and* $n \geq 2$, *we show that every fault-free edge of* Q_n *lies on a fault-free cycle of even length from 4 to* $|V| - 2f_v$ *inclusive provided* $f_e + f_v \leq n-2$. *This result is not only optimal, but also improves on the previously best known results reported in the literature.*

1 Introduction

Network topology is usually represented by a graph, $G = (V, E)$, where the set of vertices $V(G)$ represents processors, and the set of edges $E(G)$ represents links between processors. Cycles (rings) are one of the most fundamental structures for parallel and distributed computation. To find cycles of all lengths from 3 to $|V|$ is a pancyclic problem which has been investigated in a lot of interconnection networks [1, 3, 7, 8, 10, 11, 14, 18, 22]. However, the bipartite graphs contain no odd cycles. Therefore, finding cycles of all even lengths from 4 to $|V|$ is a bipancyclic problem. The concept of bipancyclicity has been extended to both of vertex-bipancyclicity [6] and edge-bipancyclicity [1]. A graph is *vertex-bipancyclic* (*edge-bipancyclic*) if every vertex (edge) lies on a cycle of every even length from 4 to $|V|$. Evidently, the edge-bipancyclic property is not only more important but also stronger than the other properties [16].

Fault-tolerance is also a desirable feature in massive parallel systems that has a relatively high probability of failure. It is useful to consider faulty networks because vertex faults and edge faults may occur simultaneously. Let f_e (respectively, f_v) denotes the number of faulty edges (respectively, faulty vertices) of the given graph. A bipartite graph G is *k-fault almost edge-bipancyclic* if every fault-free edge of G lies on a fault-free cycle of every even length from 4 to $|V| - 2f_v$ inclusive provided $f_e + f_v \leq k$. If all faulty vertices are from the same bipartite set of Q_n, such length is the best. Furthermore, each component of a network may have different reliability. Evidently, a graph G is *conditionally faulty* if each vertex is incident with at least m fault-free edges, $2 \leq m \leq \delta(G)-1$.

An n-dimensional hypercube, denoted by Q_n, has been one of the most popular interconnection networks for parallel computer systems [12]. In this paper, we consider the problem of embedding cycles in a hypercube with faulty vertices and faulty edges. This problem has received a great deal of attention in recent years [4, 5, 15, 16, 18, 20]. Previous results about edge-bipancyclic problems in faulty hypercubes briefly reviewed as follows. Considering only faulty edges (respectively, faulty vertices), Li et al. [13] (respectively, Tsai [19]) proved that Q_n is $(n-2)$-fault almost edge-bipancyclic. Considering both faulty vertices and faulty edges, Hsieh and Shen

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[9] proved that Q_n is $(n-2)$-fault almost edge-bipancyclic. Suppose that not all faulty edges are incident to the same vertex of Q_n. Xu et al. [21] proved that for $n \geq 4$, every fault-free edge of Q_n lies on a fault-free cycle of every even length from 6 to 2^n inclusive if $f_e = n-1$ and $f_v = 0$. Recently, Sun and Jou [17] further showed that for every m, Q_n is $(n-2)$-fault almost edge-bipancyclic provided $f_e \leq n-2$ ($f_v = 0$). In this paper, we generalize the results in [9, 17] by showing that for every $m \leq \min\{n - \lceil \frac{f_v}{2} \rceil, n-1\}$, Q_n is $(n-2)$-fault almost edge-bipancyclic.

The remainder of this paper is organized as follows. Section 2 introduces some definitions, notations, and properties. In Section 3, we prove the main result. The conclusions are given in Section 4.

2 Preliminaries

For graph-theoretical terminologies and notations we refer to [2]. A graph $G = (V, E)$ is an ordered pair in which V is a finite set and E is a subset of $\{(\mathbf{u}, \mathbf{v}) | (\mathbf{u}, \mathbf{v})$ is an unordered pair of $V\}$. We say that V is the vertex set and E is the edge set. We use $V(G)$ and $E(G)$ to denote the vertex set and edge set of G, respectively. For clarity, each vertex is denoted by boldface letters. Two vertices $\mathbf{u}$ and $\mathbf{v}$ are *adjacent* if $(\mathbf{u}, \mathbf{v}) \in E$. A *path* $P = \langle \mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_k} \rangle$ is a sequence of distinct vertices in which any two consecutive vertices are adjacent. For convenience, we use $\mathbf{v_1}\mathbf{v_k}$-path to denote a path P. The edge number of $k-1$ contained in the path is called the *length* of P, denoted by $l(P)$. A *cycle* in a graph G is a sequence of vertices $\langle \mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_k}, \mathbf{v_1} \rangle$, where $k \geq 3$, such that $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_k}$ are all distinct, and any two consecutive vertices are adjacent. A cycle of length k is called a k-cycle. Two cycles are *disjoint* if they do not have any vertices in common. The graph obtained by deleting all edges of E' from G is denoted by $G - E'$. We also use $G - V'$ to denote the subgraph of G induced by $V - V'$. A faulty edge (respectively, vertex) of G is an edge (respectively, a vertex) that can be deleted from G.

An n-dimensional hypercube Q_n is a bipartite graph with 2^n vertices; each vertex $\mathbf{u}$ can be distinctly labeled by an n-bit binary string, $\mathbf{u} = x_n x_{n-1} \ldots x_2 x_1$. Assume that $e = (\mathbf{u}, \mathbf{v})$ is an edge of Q_n and two vertices $\mathbf{u} = x_n \ldots x_i \ldots x_1$ and $\mathbf{v} = x_n \ldots \overline{x_i} \ldots x_1$ are joined by an edge along dimension i, where $1 \leq i \leq n$ and $\overline{x_i}$ represents the one's complement of x_i. Then e is called *an edge of dimension* i in Q_n. In the rest of this paper, $\mathbf{u}^i$ denotes the binary string $x_n \ldots \overline{x_i} \ldots x_1$ and $(\mathbf{u})_j = x_j$. The *Hamming distance*, denoted by $d_H(\mathbf{u}, \mathbf{v})$, between two vertices $\mathbf{u}$ and $\mathbf{v}$ is the number of different bits in the corresponding strings of the vertices.

The set of all edges of dimension i in Q_n is denoted by the *crossing edge set* E_i. Clearly, $|E_i| = 2^{n-1}$. For any given i of $\{1, 2, \ldots, n\}$, let Q^0_{n-1} and Q^1_{n-1} be two $(n-1)$-dimensional subcubes of Q_n induced by all vertices with the ith bit being 0 and 1, respectively. Clearly, $Q_n - E_i = Q^0_{n-1} \cup Q^1_{n-1}$. We have Q^0_{n-1} and Q^1_{n-1} being isomorphic to Q_{n-1}. It is known that Q_n is vertex-transitive and edge-transitive [12].

Throughout this paper, let F be a subset of $V(Q_n) \cup E(Q_n)$ such that F can be decomposed into two parts F_v and F_e, where F_v (respectively, F_e) consists of f_v faulty vertices (respectively, f_e faulty edges). Suppose that Q_n is decomposed into two $(n-1)$-dimensional subcubes Q^0_{n-1} and Q^1_{n-1} along some dimension j of $\{1, 2, \ldots, n\}$. In the remainder of this paper, we use f^i_v (respectively, f^i_e) to denote the number of faulty vertices (respectively, faulty edges) in Q^i_{n-1}, $i \in \{0, 1\}$. A vertex (edge) is *fault-free* if it is not faulty. Furthermore, a path (or cycle) is fault-free if it contains neither a faulty vertex nor a faulty edge.

To prove Theorem 1, the following properties of Q_n are useful. In the next lemma, an important subgraph of Q_n will help to prove the main result.

Lemma 1 [17] *There exist exactly $n-1$ disjoint cycles in Q_n of length 4 that contain an edge $(\mathbf{u}, \mathbf{v})$ in common.*

The following two lemmas show that the basis case holds.

Lemma 2 [17] *There are fault-free cycles in Q_3 whose lengths are 4,6,8 containing any fault-free edge where $f_e \leq 1$.*

Lemma 3 [19] *There are fault-free cycles in Q_3 whose lengths are 4,6 containing any fault-free edge where $f_v = 1$.*

By definition, each vertex of Q_n can be distinctly labeled by an n-bit binary string. For any two distinct vertices $\mathbf{u}$ and $\mathbf{v}$, we can easily find an index j of $\{1, 2, \ldots, n\}$ such that $(\mathbf{u})_j \neq (\mathbf{v})_j$. Clearly, we may split Q_n along the dimension j into two $(n-1)$-dimensional subcubes such that one contains $\mathbf{u}$, and the other includes $\mathbf{v}$. As a result, the next lemma can be derived easily.

Lemma 4 *For $f_v \geq 2$ faulty vertices, Q_n can be decomposed into two $(n-1)$-dimensional subcubes along some dimension j of $\{1, 2, \ldots, n\}$ such that each contains at least one faulty vertex.*

We propose a procedure to partition Q_n with $n-2$ faults. For $n \geq 4$, the procedure partition (Q_n, F) determines a dimension j according to the following two steps:

Step 1: Suppose that $f_e \geq 1$.

We may split Q_n into two $(n-1)$-dimensional subcubes by any faulty edge of dimension j. Obviously, $f_v^i + f_e^i \leq n-3$, $i \in \{0, 1\}$.

Step 2: Suppose that $f_e = 0$.

If $f_v \geq 2$, by Lemma 4, there exists some dimension j such that $f_v^i + f_e^i \leq n-3$, $i \in \{0, 1\}$. Otherwise, we may partition Q_n along any dimension j into two $(n-1)$-dimensional subcubes. Since $n \geq 4$, $f_v^i + f_e^i \leq n-3$, $i \in \{0, 1\}$.

In summary, the proposed procedure determines a j-partition of Q_n such that for $n \geq 4$, $f_v^i + f_e^i \leq n-3$, $i \in \{0, 1\}$.

3 Main result

Theorem 1 *For all $m \leq \min\{n - \lceil \frac{f_v}{2} \rceil, n-1\}$, Q_n is $(n-2)$-fault almost edge-bipancyclic provided $n \geq 2$.*

Proof: First, we need to prove that for $n \geq 2$, Q_n is $(n-2)$-fault almost edge-bipancyclic in advance. We proceed by induction on n. The induction base, depending upon $n = 2, 3$, follows from Lemmas 1-3. As our inductive hypothesis, we assume that the result holds for Q_{n-1} when $n \geq 4$. Let $e = (\mathbf{u}, \mathbf{v})$ be any fault-free edge in Q_n. Applying the procedure partition (Q_n, F), we can determine a j-partition of Q_n such that $f_v^i + f_e^i \leq n-3$, $i \in \{0, 1\}$. Since Q_n is edge-transitive, without loss of generation, we assume that $j = n$. From now on, it is necessary to construct fault-free cycles in Q_n containing e whose lengths are $4, 6, \ldots, 2^n - 2f_v$. There are two scenarios.

Case 1: Suppose that e lies in Q_{n-1}^i for some $i \in \{0, 1\}$.

Without loss of generation, assume that e lies in Q_{n-1}^0. Since $f_v^0 + f_e^0 \leq n-3$, by induction, there exist fault-free cycles in Q_{n-1}^0 containing e whose lengths are $4, 6, \ldots, 2^{n-1} - 2f_v^0$. Subsequently, we build a fault-free even k-cycle in Q_n containing e where $2^{n-1} - 2f_v^0 + 2 \leq k \leq 2^n - 2f_v$. Let C_0 be one of the fault-free longest cycles in Q_{n-1}^0 containing e. Obviously, $l(C_0) = 2^{n-1} - 2f_v^0$. Let $l_1 = l - l(C_0) - 1$. Because both l and $l(C_0)$ are even, l_1 is odd, i.e., $l_1 = 1, 3, \ldots, 2^{n-1} - 2f_v^1 - 1$. Since $l(C_0 - e) = 2^{n-1} - 2f_v^0 - 1 \geq 2^{n-1} - 2(n-3) - 1$, there are at least $2^{n-2} - (n-3) - 1$ disjoint edges in $C_0 - e$. Because $2^{n-2} - (n-3) - 1 > n-3$, there exists an edge $(\mathbf{x}, \mathbf{y})$ in $C_0 - e$ such that three edges $(\mathbf{x}, \mathbf{x}^n)$,$(\mathbf{y}, \mathbf{y}^n)$, and $(\mathbf{x}^n, \mathbf{y}^n)$ are fault-free; moreover, $d_H(\mathbf{x}^n, \mathbf{y}^n) = 1$. Because $f_v^1 + f_e^1 \leq n-3$, by induction, there exists a fault-free cycle C_1 in Q_{n-1}^1 of every even length from 4 to $2^{n-1} - 2f_v^1$ containing $(\mathbf{x}^n, \mathbf{y}^n)$. Since a cycle can be traversed backward and forward, C_1 contains fault-free $\mathbf{x}^n\mathbf{y}^n$-paths P_1 of Q_{n-1}^1 whose lengths are $1, 3, \ldots, 2^{n-1} - 2f_v^1 - 1$. Merging the paths $C_0 - (\mathbf{x}, \mathbf{y})$ and P_1 as well as the edges $(\mathbf{x}, \mathbf{x}^n)$ and $(\mathbf{y}, \mathbf{y}^n)$, the required cycle of even length k is established.

Case 2: Suppose that e belongs to the crossing edge set E_n.

Without loss of generation, assume that $\mathbf{u}$ belongs to Q_{n-1}^0. Since $f_v + f_e \leq n-2$, by Lemma 1, there exists a fault-free cycle C of length 4 that contains the edge $(\mathbf{u}, \mathbf{v})$. Write C as $\langle \mathbf{u}, \mathbf{x}, \mathbf{x}^n, \mathbf{v}, \mathbf{u} \rangle$. Clearly, $(\mathbf{u},\mathbf{x})$ is contained in Q_{n-1}^0, and $(\mathbf{v}, \mathbf{x}^n)$ is included in Q_{n-1}^1. Since $f_v^0 + f_e^0 \leq n-3$, by induction, there exist fault-free cycles C_0 in Q_{n-1}^0 containing $(\mathbf{u}, \mathbf{x})$ whose lengths are $4, 6, \ldots, 2^{n-1} - 2f_v^0$. Similarly, there exist fault-free cycles C_1 in Q_{n-1}^1 containing $(\mathbf{v}, \mathbf{x}^n)$ whose lengths are $4, 6, \ldots, 2^{n-1} - 2f_v^1$. Since a cycle can be traversed backward and forward, C_0 contains fault-free $\mathbf{ux}$-paths P_0 of Q_{n-1}^0 whose lengths are $1, 3, \ldots, 2^{n-1} - 2f_v^0 - 1$, and C_1 also includes fault-free $\mathbf{x}^n\mathbf{v}$-paths P_1 of Q_{n-1}^1 whose lengths are $1, 3, \ldots, 2^{n-1} - 2f_v^1 - 1$. Combining the paths P_0 and P_1 with the edges $(\mathbf{x}, \mathbf{x}^n)$ and $(\mathbf{u}, \mathbf{v})$, a fault-free cycle in Q_n of $\langle \mathbf{u}, P_0, \mathbf{x}, \mathbf{x}^n, P_1, \mathbf{v}, \mathbf{u} \rangle$ with even length from 4 to $2^n - 2f_v$ can be generated and e is included in it.

Since $|F| = f_v + f_e \leq n-2$, $\delta(Q_n - F) \geq 2$. And because $|E_i| = 2^{n-1} > n-2 \geq |F|$, $Q_n - F$ with $\delta(Q_n - F) \geq m$ exists for $m = 2, 3, \ldots, n-1$. By combining the above cases, Q_n is $(n-2)$-fault almost edge-bipancyclic for every m.

Conversely, assume two vertices $\mathbf{u}$ and $\mathbf{v}$ in Q_n are joined by an edge along dimension n. We can choose a set R_v consisting of f_v faulty vertices $\{\mathbf{u}^1, \mathbf{v}^2, \mathbf{u}^3, \mathbf{v}^4, \ldots, \mathbf{u}^{f_v}\}$ if f_v is odd; otherwise, $\{\mathbf{u}^1, \mathbf{v}^2, \mathbf{u}^3, \mathbf{v}^4, \ldots, \mathbf{v}^{f_v}\}$. Similarly, we can also select a set R_e comprising of $f_e = n-1-f_v$ faulty edges, $\{(\mathbf{u}^i, \mathbf{v}^i) \mid f_v + 1 \leq i \leq n-1\}$. Clearly, $f_e + f_v = n-1$ and each vertex of $Q_n - R_e - R_v$ has at least $k = \min\{n - \lceil \frac{f_v}{2} \rceil, n-1\}$ fault-free edges incident to it, that is, $\delta(Q_n - R_e - R_v) \geq k$. However, by Lemma 1, it is impossible to make a cycle of length 4 that contains the edge $(\mathbf{u}, \mathbf{v})$. As a result, Q_n is $(n-2)$-fault almost edge-bipancyclic for every $m \leq \min\{n - \lceil \frac{f_v}{2} \rceil, n-1\}$ and $n \geq 2$.

The theorem is completed.

From Theorem 1, the following proofs are straightforward.

Corollary 1 [9] *For $n \geq 2$, Q_n is $(n-2)$-fault almost edge-bipancyclic.*

Corollary 2 [17] *For every integer m, Q_n is $(n-2)$-fault almost edge-bipancyclic provided $f_e \leq n-2$ $(f_v = 0)$ and $n \geq 2$.*

4 Conclusion

In this paper, we define conditionally fault-tolerant edge-bipancyclicity for bipartite graphs with faulty vertices and faulty edges, and discuss such properties on hypercubes. We have shown that for every integer $m \leq \min\{n - \lceil \frac{f_v}{2} \rceil, n-1\}$ and $n \geq 2$, the conditionally faulty hypercube with $\delta(Q_n - F) \geq m$ is $(n-2)$-fault almost edge-bipancyclic and the result is optimal with respect to the number of faults tolerated.

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